

Islamic University of Technology (IUT)

Department of Computer Science and Engineering (CSE)

Assignment 1, Winter Semester 2024-2025

Course Title: Geometry and Differential Calculus

Course No.: MATH 4141, Section No.: SWE

Name:

Student ID.:

1. (a) Remove the first-degree terms from each of the following equations:
 - (i). $3x^2 + 4y^2 - 12x + 4y + 13 = 0$;
 - (ii). $x^2 + 3y^2 + 2x + 12y + 1 = 0$;
 - (iii). $3x^2 - 4y^2 + 6x + 24y - 135 = 0$.
- (b) For each of the following equations, determine the appropriate rotation formula so that once the rotation of axes is made the equation contains no xy -term. Also, sketch and analyze the new equations.
 - (i). $10x^2 + 12xy + 4y^2 - x - y + 10 = 0$;
 - (ii). $16x^2 + 24xy + 9y^2 - 60x + 80y = 0$;
 - (iii). $x^2 - 6xy - 5y^2 + 4x - 22 = 0$.
2. (a) Determine whether the second-degree equation $2x^2 - xy - 3y^2 - 6x + 19y - 20 = 0$ represents a pair of straight line or not. If so, then
 - (i). Separate the lines, and find their point of intersection and angle between them.
 - (ii). Find the equation of the bisectors of the angles between the lines.
 - (iii). Plot the lines in the same set of axes using different styles.
 - (iv). Find the area of the triangle formed by the lines and x -axis.

- (b) Find the value of λ for which the general equation of second degree $12x^2 + \lambda xy + 2y^2 + 11x - 5y + 2 = 0$ represents pair of straight lines. Also, find the point of intersection and the angle between them.
- (c) Find the value of k , if the lines joining the origin to the points of intersections of the curve $2x^2 - 2xy + 3y^2 + 2x - y - 1 = 0$ and $x + 2y = k$ are mutually perpendicular. Verify this conjecture graphically.
3. (a) Find the direction cosines of the line that is drawn from the origin to the point $(4, -4, 7)$.
- (b) Find the distance of the point $P(2, -1, 4)$ from the line through $Q(1, 3, -2)$ whose direction cosines are proportional to $3, 1, -2$.
- (c) Find the projection of the joining of two points $(2, -1, 3)$ and $(5, 1, 4)$ on the line through the points $(1, 0, -2)$ and $(4, 2, 1)$.
4. (a) Find and sketch the equation of the plane that passes through the points $(1, 2, 1)$, $(3, 0, 2)$, and $(0, 1, 3)$.
- (b) Find the distance between the plane found in (a) and the point $(5, -2, 3)$ and explain it geometrically.
- (c) Show that the four points $(2, -1, 3)$, $(5, 0, 1)$, $(-1, 4, 2)$, and $(2, 1, 2)$ are coplanar.
- (d) Show that the planes $3x - 2y + 5z - 7 = 0$ and $10x + 5y - 4z + 15 = 0$ are perpendicular to each other.
- (e) Find the acute angle between the planes $2x - y + 2z - 3 = 0$ and $x + 3y - z + 1 = 0$.
5. (a) Find the equation of the plane passing through the intersection line of the planes $3x - 2y + z - 5 = 0$ & $4x + y - 3z + 2 = 0$ and the point $(2, -1, 1)$.
- (b) (i). Find the equation of the plane which is parallel to the plane $5x - 3y + 2z - 7 = 0$ and at a distance of 6 units from the point $(2, -1, 3)$.

- (ii). Find the equation of the plane which is parallel to the plane $4x - 4y + 7z = 0$ and at a distance of 4 units from the point $(4, 1, -2)$.
- (c) (i). Find the equation of the plane which passes through the given points $(1, 2, -1)$ and $(4, -1, 3)$, and is perpendicular to the plane $3x + 5y - 2z + 4 = 0$.
- (ii). Find the equation of the plane which passes through the given points $(2, 2, 1)$ and $(9, 3, 6)$, and is perpendicular to the plane $2x + 6y + 6z - 9 = 0$.
6. (a) (i). Show that the lines $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ are coplanar. Also, find their point of intersection and the equation of the plane containing them.
- (ii). Show that the lines $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{-3}$ and $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$ are coplanar. Also, find their point of intersection and the equation of the plane containing them.
- (b) (i). Find the shortest distance (SD) between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$. Hence, find the equation of the SD line and the points at which it meets the given lines.
- (ii). Find the shortest distance (SD) between the lines $\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-5}{4}$ and $\frac{x-5}{2} = \frac{y-2}{3} = \frac{z-7}{5}$. Hence, find the equation of the SD line and the points at which it meets the given lines.

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